A DIRECT SUM OF INJECTIVE MODULES OVER A NOETHERIAN RING IS INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-dualizing/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The Baer criterion for injectivity through extensions from ideals. Invoked inside the counted proof (source lines 515-517) as a cross-chapter pointer. Reduces injectivity to extending maps from ideals.
- The statement that ideals of a Noetherian ring are finitely generated. Used inside the counted proof (source lines 518-520). Confines the image of the map to finitely many summands.

Source reference. Original reference injectives-lemma-criterion-baer: sources/stacks-dualizing/pinned-original-source.tex, source lines 517–517. Navigation pointer only: the counted text refers to the Stacks item injectives-lemma-criterion-baer, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let R be a Noetherian ring. A direct sum of injective modules is injective.*

Proof. Let E_i be a family of injective modules parametrized by a set I. Set $E = \bigoplus E_i$. To show that E is injective we use Injectives, Lemma injectives-lemma-criterion-baer. Thus let $\varphi : J \to E$ be a module map from an ideal of R into E. As J is a finite R-module (because R is Noetherian) we can find finitely many elements $i_1, \ldots, i_r \in I$ such that φ maps into $\bigoplus_{j=1,\ldots,r} E_{i_j}$. Then we can extend φ into $\bigoplus_{j=1,\ldots,r} E_{i_j}$ using the injectivity of the modules E_{i_j}. □

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.